Software Engineering - Project Proposal

Project Title: Task Manager Application

Description: A task management tool to organize and track work progress

Tech Stack: JavaScript (React.js + Node.js) / Python (Django)

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Introduction

In our day-to-day lives, we often have multiple things to remember and complete, such as assignments, meetings, personal work, deadlines, and other responsibilities. When these tasks are managed through memory, handwritten notes, or scattered across different applications, it becomes easy to forget important work or lose track of what needs to be done.

The Task Manager Application is proposed as a simple and organized solution to this problem. The main idea behind the project is to provide users with one place where they can create, organize, monitor, and complete their personal tasks.

The application will allow users to create an account and securely log in before accessing their tasks. Once logged in, users can create tasks with useful information such as a title, description, due date, priority, category, and completion status. They can then organize their tasks, filter and sort them according to their requirements, and mark them as completed when the work is finished.

The system will also include a dashboard that gives users a quick overview of their tasks and progress. Instead of checking every task individually, users can view summary statistics and graphical information to understand how much work is pending or completed. The system will also provide reminders when deadlines are approaching or have arrived.

The proposed application is therefore not intended to be an overly complicated productivity platform. Instead, the focus is on creating a simple, secure, reliable, and practical task management system that makes everyday task tracking easier.

Problem Statement

Managing several tasks at the same time can become difficult, especially when there are different deadlines, priorities, and categories involved. Users may forget deadlines, overlook important tasks, or find it difficult to understand their overall progress.

Traditional methods such as notebooks, sticky notes, or maintaining separate lists can also

make it difficult to update and track tasks efficiently. On the other hand, using multiple applications for different purposes can make information scattered and inconvenient to manage.

There is therefore a need for a centralized application where users can maintain their personal tasks, organize them based on priority and category, receive deadline reminders, and easily monitor their progress.

The Task Manager Application aims to address this problem by bringing these essential task-management activities together in one system.

Aim of the Project

The main aim of the project is to develop a secure and user-friendly Task Manager Application that allows individuals to efficiently create, organize, track, and complete their personal tasks from a single platform.

The application will focus on making task management simple while providing enough features to help users stay organized and aware of their progress.

Objectives

The major objectives of the proposed project are:

- To provide users with a simple way to create and manage personal tasks. ● To allow users to assign due dates, priorities, descriptions, and categories to their tasks. ● To provide filtering and sorting options for better task organization. ● To allow users to mark tasks as completed and remove tasks that are no longer required.
- To provide reminders for approaching or reached deadlines.
- To provide a dashboard showing task statistics and progress.
- To maintain separate task data for each registered user.
- To protect user accounts through authentication and secure password storage.
- To maintain task information across different user sessions.
- To maintain reliable and consistent data without unauthorized modifications.
- To support database backup and recovery.
- To maintain timestamps for important task-related activities.

These objectives are directly aligned with the functional and non-functional requirements defined for the system. The requirements specify 28 functional requirements covering account, task, organization, reminder, dashboard, storage, access, and session features.

Proposed Solution

The proposed solution is a centralized Task Manager Application in which each user has a

personal account and workspace.

A typical user journey would begin with registration and login. During registration, the system will validate the user's email format and ensure that the password meets the required strength. After successful login, the user will be taken to their task-management area.

From there, the user can create tasks by entering a title and, if required, a description. Additional information such as due date, priority, and category can be assigned to make the task easier to manage.

Users will be able to view their tasks and organize them according to their needs. For example, they may choose to see only high-priority tasks, tasks belonging to a particular category, or tasks that are still pending. They can also sort tasks based on factors such as due date or priority.

A dashboard will provide a broader picture of the user's work. It will display task statistics and graphical progress so that users can quickly understand their current task situation.

The application will also handle reminders for upcoming or reached deadlines. Behind the scenes, user and task information will be stored persistently in a database, with access controls ensuring that users can only access their own information.

Major Features

User Account Management

Users will be able to:

- Register for an account using their email and password.
- Log in using valid credentials.
- Log out securely.
- View and edit their profile.
- Change their password after verifying the current password.
- Reset a forgotten password through email verification.

Email validation and password-strength enforcement are also included in the

requirements. **Task Management**

The main part of the application will allow users to:

- Create new tasks.
- Add a mandatory task title.
- Add an optional description.
- Set a due date.
- Assign a priority.
- Assign a category.
- View task details.

- Update task priority.
- Mark tasks as completed.
- Permanently delete unnecessary tasks.

These capabilities form the core task-management functionality of the application.

Task Organization

To prevent the task list from becoming difficult to manage, users will be able to create and maintain categories.

Tasks can be grouped into categories and users can filter their task list based on status, priority, or category. Sorting options will also allow users to arrange tasks according to attributes such as due date and priority.

Reminders

The system will provide reminders when a task's due date is approaching or has arrived. This feature is intended to help users stay aware of important deadlines instead of relying entirely on memory.

Dashboard and Progress Tracking

The dashboard will provide users with a quick summary of their current tasks.

It will include:

- Task statistics.
- Completion information.
- Graphical representation of task progress.

This allows users to understand their workload at a glance rather than manually checking individual tasks.

Data Security and Privacy

Since the application stores personal task information, security will be an important part of the project.

Only authenticated users will be allowed to access task-related features. Passwords will be stored using one-way hashing rather than as plain text, and the system will ensure that one user cannot access or modify another user's tasks.

Target Users

The primary target users are individuals who need a simple way to organize and monitor their personal tasks.

This can include:

- College students managing assignments and deadlines.
- Working professionals managing daily responsibilities.
- Individuals maintaining personal to-do lists.
- Anyone who wants a centralized place to track pending and completed work.

The application is designed so that users do not require specialized technical knowledge. Basic familiarity with computers or similar devices should be sufficient to register, enter information, navigate the application, and manage tasks.

Scope of the Project

The scope of the project covers the complete basic lifecycle of personal task management.

The application will cover:

User Management

- Registration
- Login and logout
- Profile management
- Password change
- Password recovery

Task Management

- Task creation
- Task viewing
- Task editing
- Task completion
- Task deletion
- Priority management
- Due dates
- Descriptions

Organization

- Category creation
- Category editing
- Category deletion
- Task filtering
- Task sorting

Monitoring

- Deadline reminders

- Dashboard statistics
- Graphical progress tracking

Security and Reliability

- Authentication
- User-specific data access
- Password hashing
- Session timeout
- Persistent data storage
- Backup and recovery
- Activity timestamps

The defined project scope also specifies that the system should support at least 1,000 tasks per user while maintaining the required dashboard performance.

System Security and Reliability

Security will be considered throughout the development of the application rather than treated as an additional feature.

The system will ensure that task-related functions are available only after authentication. User data will remain separated, preventing unauthorized access to another user's tasks.

Passwords will not be stored directly in the database. Instead, they will be protected using a one-way hashing mechanism.

The system will also include session timeout functionality to automatically terminate inactive sessions. In addition, important task activities such as creation, editing, completion, and deletion will have timestamps recorded.

To improve reliability, the database will support manual backup and restoration. The system requirements also specify that submitted task operations should not be lost and that task information should not change unless the authorized user performs the relevant action.

Performance Requirements

The application is expected to provide a smooth experience during normal usage.

One of the main measurable performance goals is that the dashboard should load within 3 seconds under normal conditions.

The system should also be capable of handling at least 1,000 tasks for a single user without causing the dashboard performance to go beyond the specified threshold.

These requirements give the project clear and measurable quality targets instead

of evaluating performance only on the basis of whether the application works.

Benefits of the Proposed System

The proposed Task Manager Application can provide several practical benefits:

- **Better organization:** Users can keep their tasks in one centralized location. ● **Improved deadline awareness:** Reminders can help users remember approaching deadlines.
- **Easy prioritization:** Priority levels help users identify tasks that require more attention.
- **Simpler task tracking:** Users can filter and sort tasks instead of manually searching through them.
- **Progress visibility:** The dashboard gives users a quick understanding of their completed and pending work.
- **Data security:** Individual accounts keep users' information separate. ● **Data availability:** Persistent storage ensures that tasks remain available across sessions.
- **Reliability:** Backup, recovery, and data-integrity requirements help protect stored information.

Feasibility

The proposed project is feasible because its requirements are clearly defined and the system is focused on a manageable set of core features.

From a technical perspective, the application consists of commonly implemented components such as user authentication, database operations, task management, filtering, dashboards, and notifications.

From an operational perspective, the system is designed around simple interactions that users are already familiar with, such as registration, login, forms, task lists, and dashboards.
From a project perspective, the system can also be developed incrementally. Core features such as authentication and task creation can be implemented first, followed by organization, reminders, dashboard functionality, and security/reliability improvements.

Conclusion

The Task Manager Application is proposed as a practical solution for people who need a simple and organized way to manage their personal tasks and deadlines.

The project brings commonly required task-management activities into one application, including task creation, prioritization, categorization, filtering, reminders, completion tracking, and progress monitoring. At the same time, it considers important software engineering qualities such as security, performance, data integrity, scalability, backup,

recovery, and traceability.

The project's requirements provide clear measurable targets, including a 3-second dashboard loading requirement, secure password storage, prevention of cross-user data access, support for at least 1,000 tasks per user, and timestamp tracking for important task activities.

Overall, the goal is to build a system that is simple enough for everyday use but structured enough to demonstrate proper software engineering principles. The project therefore combines a real-world problem with a well-defined software solution and provides a strong foundation for designing, developing, testing, and evaluating the Task Manager Application.